Alexander Loker

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Education

University of Waterloo *2025 – Present*
Bachelor of Applied Science in Mechatronics Engineering – 3.7 GPA *Waterloo, ON, Canada*

Skills

Tools: Altium, LTspice, PSpice, SolidWorks, AutoCAD, Soldering, 3D Printing, Oscilloscope
Electrical: Schematic Capture, PCB Layout, Mixed-Signal, Analog, Power Distribution & Management, DAC, MCU, RasPi,
Software & Protocols: C++, Python, Git, Linux, PWM, DMX512, I2C, SPI, UART, USB

Work Experience

WET Design *May – Aug '26*
Product Designer & Engineer *Sun Valley, CA, USA*

- Built a handheld **Digital Multiplex** lighting rig tester from the ground up, replacing the external option saving **$900** per unit while adding fault protection and an externally programmable layer.
- Designed a custom **RasPi CM4** carrier board in **Altium** from schematic to PCB layout, integrating a WET RS-485 port, battery management, and power distribution.
- Safely integrated a **USB-C, BQ charger IC, buck-boost IC,** and CM4 3.3V rail, using low-impedance routing and proper thermal balancing enabling the use of a 3.7V 6Ah power supply.
- Iterated the enclosure around software and PCB revisions while customizing UX/UI in **SolidWorks** and **PyGame** to accommodate a modular script, on-site safe, and one-handed design.
- Managed the team's Weigl ProCommander HXi integration for a fountain rig: Wiring and documenting the control box across DMX, analog and digital I/O, eventually training 4 engineers on the platform.

Projects

Digital Audio Player *Sep – Present '26*

- Developed a **mixed-signal** 55x55 mm **PCB** from scratch for convenient & high-quality lossless listening: Mixing **analog** audio design with power distribution, battery charging, **ESP32-S3** control, and a physical UI.
- Designed a low-noise **DAC** back end including 4x precision **Op-Amps** for an ultra-low noise output: I/V Conversion, to differential summing with a 2nd-order **MFB Butterwoth low-pass**, to a unity-gain driver.
- Ensured proper bypass, decoupling, and low-pass filtering on both sensitive analog and digital rails via **LTspice** & **PSpice** simulation.
- Protected the analog chain from 4 switching regulators (0.6 - 3 MHz) through via-fencing, shielded edge-of-board inductors, and two unbroken ground planes. Limiting **EMI** and therefore noise into the audio output.

Companion Robot *Jan – Apr, Sep – Present '26*

- Led end-to-end hardware design of an autonomous companion robot spanning mechanical design (**SolidWorks**), electrical components and routing.
- Replaced off-the-shelf parts with a custom battery management and power distribution PCB, designed in **Altium** and verified in **WEBENCH**, for reliable and safe power management.
- Implemented a micro-motor tri-omniwheel drivetrain on a **Pi Pico** via **SolidWorks** modeling, 3D printing, and waffle-board soldering.
- Compacted robot volume by **26%** while embedding servos, microphones, and a camera module directly into the shell by using compact interlocking joints.

Under-voltage Lockout Circuit – Biotron Exoskeleton Team *Feb – Mar '26*

- Designed and verified a **UVLO** circuit in **LTspice**, and transformed the schematic into a PCB in **Altium**.
- Created under-voltage lockout logic via **Schmitt trigger hysteresis** and controlled current flow through a **dual-MOSFET switch system** ensuring voltage lockout at **24.1V**.